Sundar Reddy R

Technical Lead | Backend systems

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Technical Skills

Frameworks: Java (Quarkus, Springboot), Python (FastAPI), SQL, Dart (Flutter)
Libraries: Hibernate ORM+JPA, Blaze persistence, Liquibase, Pandas, SqlAlchemy, Pydantic, Boto3
Databases: MySQL, ClickHouse, Opensearch, Oracle SQL
Cloud & DevOps: AWS(EC2, Cognito, IAM, S3, CloudFront, SNS, SQS, VPC, Bedrock), Github CI/CD, Docker Compose, ECR, Postman
Backend: REST APIs, Data Modeling, Transactions, DB Optimization & Scaling, Authentication & Authorization, Multi-tenancy, Event-driven architecture, Concurrency, Caching
Agentic Engineering: RAG, Harness Enginerring, Context Mangement, Spec-driven development, MCP, Codex/Claude code, AI-assisted Code review

Experience

Trumetric.AI — *Bengaluru, IN*

Technical Lead — ***Apr 2026 - Present***

Trusearch - Enterprise Search for CMS

- Took over Trusearch's agent-first codebase and led a team of 5 to bring it to production readiness, owning architecture, client requirements, deployment decisions, and sprint delivery.
- **Team Management:** Streamlined feature delivery through sprint planning, architecture guidance and blocker resolution. Aligned developers, QA, and DevOps across testing and deployment. Led client discussions throughout onboarding & go-live.
- **Harness Engineering**: Tackled agentic bottlenecks like oversized PRs and context loss problems through spec-driven development, PR limits, and persistent repository memory. Automated agent reviews with human in the loop oversight. Mandated walkthroughs & QA checks to validate code quality and developer understanding.
- **RAG:** Built AI Overview with GPT 5.6-luna to summarize top-k search results with source citations and conversion-focused CTAs. Also evaluated a self-hosted open-weight alternative, Granite 4.1-8B, using vLLM on EC2.
- **MCP Integration:** Built a tenant-scoped MCP server to let clients query their analytics and search their content daily within the Claude conversation.
- **Architecture:** Designed a privacy-first architecture with the search engine and indexed content hosted in client infrastructure, alongside a centralized platform for tenants, plans, and usage limits. Led a low-code widget builder serving JS/CSS through S3 + CloudFront for website integration, and guided marketing on Generative Engine Optimization (GEO).
- **Search Engineering:** Led one-click Integration that indexes ~10000 documents in 3 minutes & initializes widgets to make Search available instantly. Built hybrid retrieval combining BM25, HNSW dense vectors, and neural sparse search, supporting typeahead and suggestions at **p95 <300 ms**. Addressed a 10x latency after Re-ranking through Caffiene L1 caching.

  Stack: Java (Quarkus), Opensearch, Redis, Postgres, Sqllite, S3, Cognito, Cloudfront, EC2, Docker compose

Livine (Poultry ERP) — *Bengaluru, IN*

Technical Lead — ***Apr 2025 – Apr 2026***

Grower App

- Owned design and delivery of India's first Smart Grower App (Flutter+Quarkus) aimed at automating Breeder operations, reducing manual labour and providing Real-time analytics & projections.
- **Client Engagement:** Led the entire process from requirement gathering to on-site implementation, from guiding users to troubleshooting issues and all further client communication & feedback loop.
- **Data processing:** Phone sits in your pocket and reads values from a scale over BLE protocol & uses ***Sliding-window Standard Deviation*** approach to capture stable weights automatically (~200ms interval). Eliminated the need for a separate person who records weights on paper.

- **Offline operation & Concurrency:** Designed a fail-proof offline first weighing to make up for poor farm connectivity and speed up processing (~3s per bird). Batch-wise Syncing once good connectivity is detected. Enabled 10+ concurrent users to process upto ***50k birds per day*** per farm.
- **Predictive Analytics:** Solved a real bottleneck with Instant analytics as soon as data is synced, replacing a manual process that took until evening or next day. Built a Gompertz-based growth model using SciPy curve fitting to generic farm-specific weekly body weight targets.

  Stack: Java (Quarkus), Hibernate, Blaze Persistence, Flutter, Mysql, Python (FastAPI), Pandas, Scipy

Finance Module

- Led a team of 3 to build the ERP's Accounting module from scratch, owning client requirements, solution design, sprint planning and delivery.
- **Event-driven:** Integrated the ERP with the accounting module using AWS SNS/SQS to asynchronously consume Purchase/Sale events to generate corresponding Journal entries. Eliminates manual re-entry into another software like Tally.
- **Pdf Extraction:** Built a cost-effective alternative to third-party statement processing software using ***AWS Textract+Claude on Bedrock***. Statement extraction & automatic Payment/Receipt entries creation is followed.
- **Webhooks:** Designed a pluggable provider ***SPI*** to support payment webhooks, automating Receipt entries from providers like Greenorange, Razorpay etc., with Signature verification & Idempotency.
- Led hiring efforts across multiple roles during this time.

  Stack: Java (Quarkus), Hibernate, Blaze Persistence, Mysql, Cognito, SNS, SQS, AWS Textract, Bedrock

Software Developer ***Apr 2022 – Mar 2025***

- **Multi-tenancy:** Solved a painful scaling problem in managing 300+ business reports across multiple clients by building a multi-tenant React + Quarkus platform, enabling client-specific module/category activation and one-click updates across tenants.
- **OLAP Migration:** Relieved pressure on MySQL by offloading read-heavy reporting to ClickHouse, cutting report latency from >1 minute to 2–3 seconds and infrastructure costs by an estimated 40–60%.
- **DB Performance**: Tried multiple ways to optimize querying with Views, Pre-aggregated tables with scheduled jobs, partioning etc., Eventually also came up with a FastAPI Microservice using Pandas for advanced reporting.
- Built 300+ business reports & dashboards over a span of 1.5 years.

Lumen Technologies *Bengaluru, IN*

Software Development Consultant - I ***Jan 2022 – Apr 2022***

- Led application modernization by migrating users to Azure B2C for identity management, leveraging Azure IAM and Salesforce Marketing Cloud for all types of customer interactions.
- Involved in DevOps processes such as building CI/CD pipelines using Jenkins and setting up Splunk to enhance application monitoring & debugging capabilities.

Software Development Engineer (SDE) ***Sep 2020 – Dec 2021***

- Developed RESTful services for a Customer Profile Management application using Spring Boot, integrating Kafka messaging and Oracle SQL; APIs were critical in the Authentication & Authorization process of many customer-facing applications.

  Stack: Java (Springboot), Oracle Sql, Azure B2C, Kafka, Postman

Education

Manipal Institute of Technology *Manipal, KA*

B.Tech. in ECE — ***CGPA: 8.0*** *2016 – 2020*